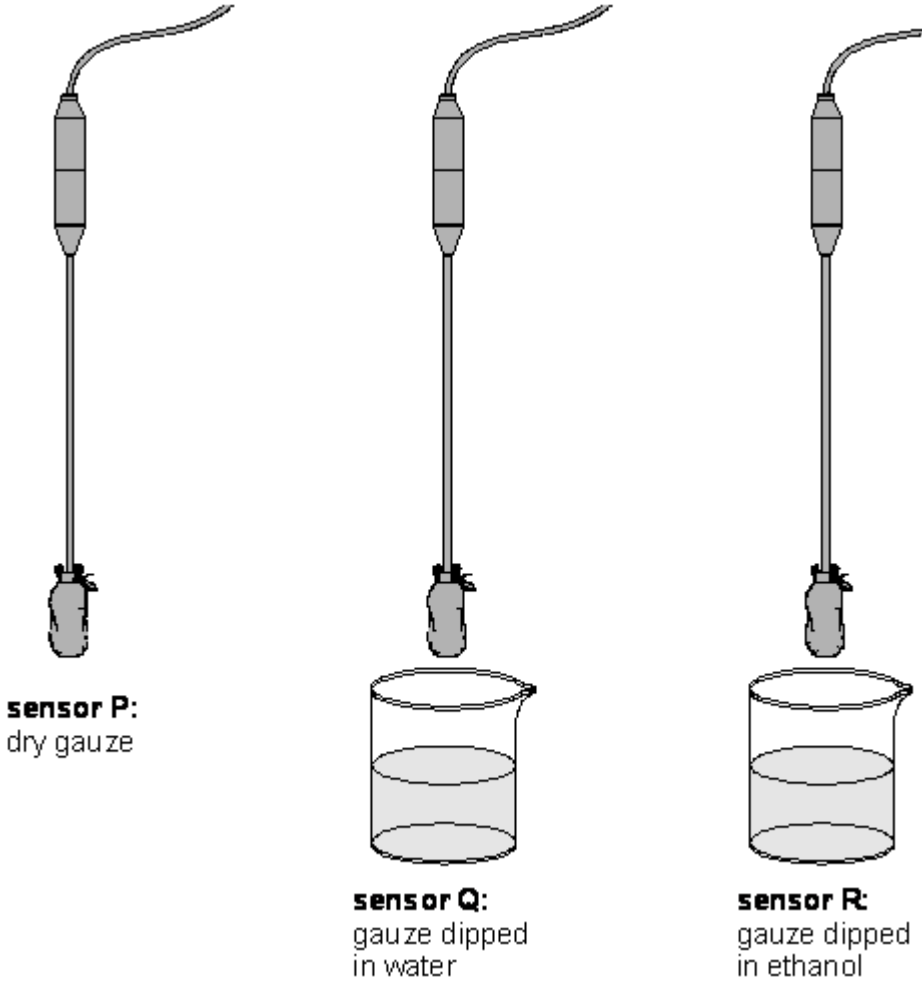
Q1. Kala recorded temperatures using a datalogger and three temperature sensors, P, Q and R. The ends of the sensors were covered with gauze.
P was kept dry.
Q was dipped in water for 2 seconds and then taken out.
R was dipped in ethanol for 2 seconds and then taken out.



Every five minutes, the datalogger recorded the temperatures of the sensors.

time, in min	temperature of sensor P, in °C	temperature of sensor Q, in °C	temperature of sensor R, in °C
0	20	20	20
5	20	18	16
10	20	17	12
15	20	17	20

(a) What was the temperature of the room at the beginning of Kala's experiment?

..... °C

1 mark

(b) Describe how the temperatures of sensor Q and sensor R changed.

sensor Q .....

.....

sensor R .....

.....

2 marks

(c) The next day the gauze on sensor Q felt dry. What had happened to the water on the gauze?

.....

1 mark

Maximum 4 marks

Mark schemes

Q1.

(a) 20

1 (L3)

(b) sensor Q: any **one** from

- it goes down
- it goes to 17°C

accept 'it levels out at 17°C'
*do **not** accept 'it levels out'*

1 (L4)

sensor R: any **one** from

- it goes down and comes up
- it goes to 12°C and back up to 20°C

accept 'it comes back up'

1 (L4)

(c) it had evaporated

accept 'it had gone into the air'
'it had dried up' is insufficient

1 (L4)

[4]